



PARSHVANATH CHARITABLE TRUST'S

A. P. SHAH INSTITUTE OF TECHNOLOGY

Department of Information Technology

(NBA Accredited)



Semester: VII

Academic Year: 2026-27

Class / Branch: BE IT-B

Subject: Recent Open Source Lab

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EXPERIMENT NO. 05

Aim: To study open-source database management tools like Cassandra, MySQL, MariaDB and understand the steps for contributing in Cassandra.

```
apsit@apsit-HP-280-G3-MT:~$ free -h
              total        used        free      shared  buff/cache   available
Mem:           7.6Gi       5.5Gi         400Mi        258Mi         1.7Gi         1.6Gi
Swap:          14Gi       1.1Gi          13Gi
apsit@apsit-HP-280-G3-MT:~$ sudo docker exec -it my-cassandra cqlsh

ATTENTION: All commands will be saved to history file: /root/.cassandra/cqlsh_history
This may include sensitive information such as passwords.
To disable history, use --disable-history or set 'disabled = true' in the [history] section of cqlshrc.
See https://cassandra.apache.org/doc/latest/tools/cqlsh.html for more information.

Connected to Test Cluster at 127.0.0.1:9042
[cqlsh 6.2.0 | Cassandra 5.0.8 | CQL spec 3.4.7 | Native protocol v5]
Use HELP for help.
cqlsh> CREATE KEYSPACE labdemo
WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 1};

cqlsh> CREATE KEYSPACE labdemo WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 1};
AlreadyExists: Keyspace 'labdemo' already exists
cqlsh>
cqlsh>
cqlsh> CREATE KEYSPACE labdemo1 WITH replication = {'class': 'SimpleStrategy', 'replication_factor': 1};
cqlsh> use labdemo1;
cqlsh:labdemo1> CREATE TABLE faculty (
...     facultyId int PRIMARY KEY,
...     name text,
...     dept text
... );
cqlsh:labdemo1> INSERT INTO faculty (facultyId, name, dept) VALUES (1, 'Clark', 'Comp');
cqlsh:labdemo1> INSERT INTO faculty (facultyId, name, dept) VALUES (2, 'Dave', 'IT');
cqlsh:labdemo1> INSERT INTO faculty (facultyId, name, dept) VALUES (3, 'Eve', 'IT');
cqlsh:labdemo1> SELECT * FROM faculty;

 facultyid | dept | name
-----
1 | Comp | Clark
2 | IT   | Dave
3 | IT   | Eve
(3 rows)
cqlsh:labdemo1>
```

Conclusion: study open-source database management tools like Cassandra, MySQL, MariaDB and understand the steps for contributing in Cassandra.